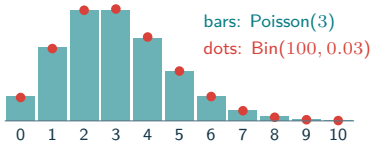
PROBABILITY LADDER 7 OF 9

Waiting times: geometric and Poisson

The geometric law times the first success, and the Poisson law counts rare events, each the limit of simpler binomial pictures.

1 / THE RULE AND THE PICTURE

When trials are many and successes rare, the binomial count settles to $e^{-\lambda}\lambda^k/k!$



If a count has mean λ and events occur independently at random, then

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad E[X] = \operatorname{Var} X = \lambda$$

It is the limit of $\mathrm{Bin}(n, \lambda/n)$ as $n \to \infty$, and already $\mathrm{Bin}(100, 0.03)$ is close to $\mathrm{Poisson}(3)$.

2 / A WORKED POISSON EXAMPLE

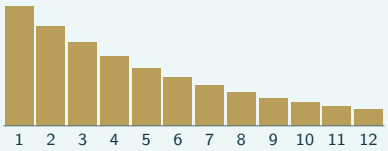
A switchboard receiving 3 calls an hour on average

calls k	Poisson (3)	$\mathrm{Bin}(100, 0.03)$
0	0.0498	0.0476
1	0.1494	0.1471
2	0.2240	0.2252
3	0.2240	0.2275
4	0.1680	0.1706
5	0.1008	0.1013
6	0.0504	0.0496

The probability of no call in an hour is $e^{-3} = 0.0498$, and of five or more calls $1 - (0.0498 + 0.1494 + 0.2240 + 0.2240 + 0.1680) = 0.185$.
The probabilities of 2 and 3 calls are equal because $\lambda = 3$.

3 / THE GEOMETRIC DISTRIBUTION / WAITING FOR A SIX

The first success on trial k has probability $(1-p)^{k-1}p$



For a die, $p = \frac{1}{6}$ and the first six comes at roll k with probability $\frac{1}{6}\left(\frac{5}{6}\right)^{k-1}$: $0.167, 0.139, 0.116, \ldots$.
The mean wait is $1/p = 6$ rolls, and the chance of a six within six rolls is $1 - (\frac{5}{6})^6 = 0.665$, with a median wait of 4 rolls since $1 - (\frac{5}{6})^4 = 0.518$.

4 / NO MEMORY / THE EXPONENTIAL WAITING TIME

Having waited without success does not shorten the remaining wait

For the geometric law $P(X > m + n \mid X > m) = P(X > n)$, because each trial starts afresh.
In continuous time the gap T between Poisson events at rate λ satisfies $P(T > t) = e^{-\lambda t}$, with mean $1/\lambda$; for $\lambda = 3$ per hour the mean gap is 20 minutes and the median $\ln 2/3 = 0.231$ hours, about 14 minutes.
Radioactive decay is of this kind, and its half-life is $\ln 2$ divided by the decay rate.

5 / WHICH LAW TO USE

Match the question to the story

law	counts	mean	variance
binomial	successes in n trials	np	$np(1-p)$
geometric	trials until the first success	$1/p$	$(1-p)/p^2$
Poisson	events in a fixed interval	λ	λ

The Poisson law fits phone calls, typing errors, goals in a football match or radioactive counts, wherever events are rare in each small moment and independent of one another.

ASIDE / THE LIMIT

Writing $(1 - \lambda/n)^n \to e^{-\lambda}$ and $\binom{n}{k}/n^k \to 1/k!$ turns the binomial formula into the Poisson formula.

LINKS TO OTHER SHEETS

The binomial starting point is *The binomial distribution*; the next sheet is *The normal distribution and the central limit theorem*.

Convention. λ is the mean number of events in the interval; p the success probability; T the waiting time between events.